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Lem serie fourier derivada

Lema 1. *Sea $f\colon \mathbb{T} \to \mathbb{C}$ en $\mathcal{C}^m(\mathbb{T})$ para algún $m \in \mathbb{N}$, entonces*

$$\forall n \in \mathbb{Z} \setminus \{0\} : \widehat{f^{(m)}}(n) = (2\pi i n)^m \, \widehat{f}(n).$$

Por tanto, $\lim\limits_{|n|\to\infty} |n|^m \left|\widehat{f}(n)\right| = 0.$

***Demostración*:** Vamos a demostrar el caso $m = 1$, el resto se demuestra por inducción. Sea $f \in \mathcal{C}^1(\mathbb{T})$, entonces por integración por partes tenemos que $\forall n \in \mathbb{Z} \setminus \{0\}$:

$$\widehat{f'}(n) = \int_0^1 f'(x)\, e^{-2\pi i n x}\,\mathrm{d}x \overset{\overset{\text{IPP}}{\downarrow}}{=} f(x)\, e^{-2\pi i n x}\Big|_0^1 + 2\pi i n \int_0^1 f(x)\, e^{-2\pi i n x}\,\mathrm{d}x = 2\pi i n\, \widehat{f}(n).$$

Para la segunda parte, de la relación $\widehat{f^{(m)}}(n) = (2\pi i n)^m \, \widehat{f}(n)$ obtenemos

$$\left|\widehat{f}(n)\right| = \frac{\left|\widehat{f^{(m)}}(n)\right|}{|2\pi i n|^m} = \frac{\left|\widehat{f^{(m)}}(n)\right|}{(2\pi)^m \, |n|^m} \implies |n|^m \left|\widehat{f}(n)\right| = \frac{\left|\widehat{f^{(m)}}(n)\right|}{(2\pi)^m}.$$

Como $f^{(m)} \in \mathcal{L}(\mathbb{T})$ por ser continua en un compacto, por el Lema de Riemann-Lebesgue sabemos que $\widehat{f^{(m)}}(n) \xrightarrow{|n|\to\infty} 0$, de donde se sigue que

$$\lim_{|n|\to\infty} |n|^m \left|\widehat{f}(n)\right| = \lim_{|n|\to\infty} \frac{\left|\widehat{f^{(m)}}(n)\right|}{(2\pi)^m} = 0.$$

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Referenciado en

- `prop-clase-ck-velocidad-convergencia-uniforme-fourier`

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